ROCSCIENCE

SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

Table of Contents

Project Summary 3
General Settings 4
Analysis Options 5
Groundwater Analysis 6
Random Numbers 7
Surface Options 8
Seismic Loading 9
Materials 10
Global Minimums 11
Method: bishop simplified 11
Method: spencer 11
Valid and Invalid Surfaces 12
Method: bishop simplified 12
Method: spencer 12
Error Code Descriptions 12
Slice Data 13
Global Minimum Query (bishop simplified) - Safety Factor: 1.1169 13
Global Minimum Query (spencer) - Safety Factor: 1.11722 14
Interslice Data 15
Global Minimum Query (bishop simplified) - Safety Factor: 1.1169 15
Global Minimum Query (spencer) - Safety Factor: 1.11722 16
Entity Information 17
Water Table 17
External Boundary 17

Slide2 Analysis Information

SLIDE - An Interactive Slope Stability Program

Project Summary

Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.852s
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	30
Tolerance:	0.0001
Maximum number of iterations:	50
Check malpha < 0.2:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	rand

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	10
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

soil	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18.82 kN/m3
Cohesion	41.65 kPa
Phi	15 °
Water Surface	Water Table
Hu Type	Custom
Hu	1
Specify alternate strength type above water surface	No

Global Minimums

Method: bishop simplified

FS	1.116900
Center:	27.747, 45.404
Radius:	31.927
Left Slip Surface Endpoint:	18.001, 15.001
Right Slip Surface Endpoint:	57.931, 35.000
Resisting Moment:	97984.2 kN-m
Driving Moment:	87728.9 kN-m
Total Slice Area:	378.796 m2
Surface Horizontal Width:	39.9301 m
Surface Average Height:	9.48648 m

Method: spencer

FS	1.117220
Center:	27.731, 45.399
Radius:	31.917
Left Slip Surface Endpoint:	18.001, 15.001
Right Slip Surface Endpoint:	57.906, 35.000
Resisting Moment:	97887.1 kN-m
Driving Moment:	87617 kN-m
Resisting Horizontal Force:	2561.04 kN
Driving Horizontal Force:	2292.34 kN
Total Slice Area:	378.192 m2
Surface Horizontal Width:	39.9051 m
Surface Average Height:	9.47727 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	2117
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	1889
Number of Invalid Surfaces:	228

Error Codes

Error Code -108 reported for 11 surfaces
Error Code -111 reported for 217 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.1169

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.331	16.0602	-16.5269	soil	41.65	15	41.6408	46.5086	24.4219	6.28948	18.1324	12.0661	5.77661
2	1.331	47.4059	-14.0498	soil	41.65	15	44.0245	49.1709	46.6337	18.5652	28.0685	35.6165	17.0513
3	1.331	77.2269	-11.5994	soil	41.65	15	46.2312	51.6356	67.5108	30.2438	37.267	58.0214	27.7776
4	1.331	105.567	-9.17024	soil	41.65	15	48.2696	53.9123	87.1056	41.3426	45.763	79.3134	37.9708
5	1.331	132.461	-6.75767	soil	41.65	15	50.1464	56.0085	105.462	51.8749	53.5868	99.5196	47.6447
6	1.331	157.934	-4.3571	soil	41.65	15	51.867	57.9303	122.61	61.8507	60.7589	118.658	56.807
7	1.331	182.003	-1.96419	soil	41.65	15	53.4356	59.6822	138.574	71.2768	67.2971	136.741	65.4646
8	1.331	204.678	0.425295	soil	41.65	15	54.8549	61.2674	153.37	80.157	73.2131	153.777	73.6203
9	1.331	225.962	2.81552	soil	41.65	15	56.127	62.6882	167.008	88.4922	78.5157	169.768	81.276
10	1.331	245.849	5.21067	soil	41.65	15	57.7484	64.4992	179.443	94.1691	85.2742	184.71	90.5406
11	1.331	264.328	7.615	soil	41.65	15	59.7345	66.7175	190.606	97.0533	93.5528	198.592	101.539
12	1.331	281.377	10.0329	soil	41.65	15	61.5532	68.7488	200.512	99.3778	101.134	211.402	112.024
13	1.331	296.969	12.4691	soil	41.65	15	63.2025	70.5909	209.14	101.132	108.008	223.116	121.984
14	1.331	311.065	14.9284	soil	41.65	15	64.679	72.24	216.463	102.3	114.163	233.707	131.407
15	1.331	323.619	17.4163	soil	41.65	15	65.9781	73.6909	222.442	102.863	119.579	243.139	140.276
16	1.331	334.569	19.9386	soil	41.65	15	67.0936	74.9368	227.027	102.8	124.227	251.366	148.566
17	1.331	343.843	22.502	soil	41.65	15	68.0173	75.9685	230.158	102.079	128.079	258.334	156.255
18	1.331	351.351	25.1139	soil	41.65	15	68.7392	76.7748	231.754	100.667	131.087	263.974	163.307
19	1.331	356.981	27.783	soil	41.65	15	69.2465	77.3414	231.721	98.5196	133.202	268.204	169.685
20	1.331	360.598	30.5195	soil	41.65	15	69.523	77.6502	229.938	95.5837	134.354	270.922	175.338
21	1.331	362.033	33.3356	soil	41.65	15	69.5484	77.6786	226.253	91.7932	134.46	272	180.207
22	1.331	361.073	36.2463	soil	41.65	15	69.2965	77.3973	220.476	87.065	133.411	271.279	184.214
23	1.331	355.082	39.2702	soil	41.65	15	68.3771	76.3704	210.871	81.2932	129.578	266.778	185.485
24	1.331	329.435	42.4313	soil	41.65	15	65.0674	72.6738	188.029	72.2473	115.782	247.509	175.262
25	1.331	297.077	45.7618	soil	41.65	15	60.9952	68.1255	160.559	61.751	98.8081	223.198	161.447
26	1.331	260.572	49.3059	soil	41.65	15	56.5693	63.1822	129.99	49.6309	80.3592	195.772	146.141
27	1.331	218.96	53.1287	soil	41.65	15	51.7009	57.7447	95.5772	35.511	60.0662	164.508	128.997
28	1.331	170.734	57.3327	soil	41.65	15	46.2495	51.6561	56.1442	18.8008	37.3434	128.276	109.475
29	1.331	113.251	62.0989	soil	41.65	15	39.7113	44.3535	10.0898	0	10.0898	85.0879	85.0879
30	1.331	40.8837	67.8166	soil	41.65	15	28.1171	31.404	-38.2385	0	-38.2385	30.7175	30.7175

Global Minimum Query (spencer) - Safety Factor: 1.11722

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.33017	16.0325	-16.503	soil	41.65	15	45.4475	50.7749	40.3372	6.28257	34.0546	26.8725	20.5899
2	1.33017	47.3244	-14.027	soil	41.65	15	48.2357	53.8899	64.2248	18.5449	45.6799	52.1742	33.6293
3	1.33017	77.0943	-11.5774	soil	41.65	15	50.6503	56.5875	85.9584	30.2108	55.7476	75.5822	45.3714
4	1.33017	105.386	-9.14923	soil	41.65	15	52.7338	58.9153	105.733	41.2976	64.4352	97.2397	55.9421
5	1.33017	132.234	-6.73752	soil	41.65	15	54.5202	60.9111	123.702	51.8185	71.8835	117.261	65.4426
6	1.33017	157.663	-4.33777	soil	41.65	15	56.0366	62.6052	139.989	61.7835	78.2053	135.738	73.9548
7	1.33017	181.691	-1.94563	soil	41.65	15	57.3045	64.0217	154.692	71.1993	83.4923	152.745	81.5457
8	1.33017	204.328	0.443114	soil	41.65	15	58.3419	65.1807	167.888	80.0698	87.8179	168.339	88.2691
9	1.33017	225.575	2.83263	soil	41.65	15	59.1629	66.098	179.637	88.3959	91.2409	182.564	94.1683
10	1.33017	245.427	5.22709	soil	41.65	15	60.2999	67.3682	190.072	94.09	95.982	195.588	101.498
11	1.33017	263.873	7.63077	soil	41.65	15	61.7897	69.0327	199.162	96.9686	102.194	207.441	110.472
12	1.33017	280.892	10.0481	soil	41.65	15	63.0695	70.4625	206.818	99.2881	107.529	217.993	118.705
13	1.33017	296.455	12.4836	soil	41.65	15	64.1455	71.6646	213.053	101.037	112.016	227.254	126.217
14	1.33017	310.525	14.9423	soil	41.65	15	65.0222	72.6441	217.872	102.201	115.671	235.225	133.024
15	1.33017	323.054	17.4296	soil	41.65	15	65.7025	73.4042	221.269	102.761	118.508	241.896	139.135
16	1.33017	333.982	19.9513	soil	41.65	15	66.1877	73.9462	223.225	102.694	120.531	247.252	144.558
17	1.33017	343.237	22.5141	soil	41.65	15	66.4773	74.2698	223.709	101.971	121.738	251.264	149.293
18	1.33017	350.727	25.1255	soil	41.65	15	66.5693	74.3725	222.679	100.556	122.123	253.898	153.342
19	1.33017	356.342	27.7941	soil	41.65	15	66.4597	74.2501	220.072	98.407	121.665	255.104	156.697
20	1.33017	359.946	30.53	soil	41.65	15	66.1427	73.896	215.813	95.4698	120.343	254.821	159.351
21	1.33017	361.37	33.3456	soil	41.65	15	65.6099	73.3007	209.8	91.6784	118.122	252.973	161.294
22	1.33017	360.404	36.2557	soil	41.65	15	64.8499	72.4516	201.903	86.9499	114.953	249.463	162.513
23	1.33017	354.554	39.2791	soil	41.65	15	63.5502	70.9995	190.712	81.1783	109.534	242.688	161.51
24	1.33017	329.102	42.4397	soil	41.65	15	60.3008	67.3693	168.151	72.1654	95.9856	223.29	151.124
25	1.33017	296.775	45.7695	soil	41.65	15	56.5405	63.1682	141.979	61.6721	80.307	200.059	138.387
26	1.33017	260.306	49.3131	soil	41.65	15	52.6953	58.8722	113.83	49.5559	64.2743	175.122	125.566
27	1.33017	218.736	53.1353	soil	41.65	15	48.7636	54.4797	83.322	35.4409	47.8811	148.352	112.911
28	1.33017	170.559	57.3387	soil	41.65	15	44.7508	49.9964	49.8861	18.7367	31.1494	119.696	100.959
29	1.33017	113.135	62.1042	soil	41.65	15	40.3771	45.1101	12.9131	0	12.9131	89.1856	89.1856
30	1.33017	40.842	67.8213	soil	41.65	15	31.5965	35.3003	-23.6975	0	-23.6975	53.8096	53.8096

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.1169

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	18.0011	15.0008	0	0	0
2	19.3321	14.6058	65.0686	0	0
3	20.6631	14.2727	139.198	0	0
4	21.9942	13.9995	219.175	0	0
5	23.3252	13.7847	302.137	0	0
6	24.6562	13.6269	385.514	0	0
7	25.9872	13.5255	466.983	0	0
8	27.3182	13.4799	544.43	0	0
9	28.6492	13.4898	615.926	0	0
10	29.9802	13.5552	679.698	0	0
11	31.3112	13.6766	734.78	0	0
12	32.6422	13.8546	780.367	0	0
13	33.9732	14.09	815.077	0	0
14	35.3042	14.3844	837.644	0	0
15	36.6352	14.7392	846.917	0	0
16	37.9662	15.1567	841.857	0	0
17	39.2972	15.6396	821.542	0	0
18	40.6282	16.191	785.169	0	0
19	41.9592	16.8148	732.073	0	0
20	43.2902	17.5161	661.744	0	0
21	44.6212	18.3007	573.861	0	0
22	45.9522	19.1762	468.345	0	0
23	47.2832	20.152	345.438	0	0
24	48.6142	21.2403	206.964	0	0
25	49.9453	22.457	64.7907	0	0
26	51.2763	23.8239	-73.4898	0	0
27	52.6073	25.3716	-199.39	0	0
28	53.9383	27.1462	-300.187	0	0
29	55.2693	29.2221	-355.176	0	0
30	56.6003	31.7358	-327.684	0	0
31	57.9313	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.11722

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	18.0011	15.0008	0	0	0
2	19.3313	14.6067	76.3476	19.7115	14.4766
3	20.6615	14.2744	161.85	41.7866	14.4766
4	21.9917	14.0019	252.645	65.2281	14.4765
5	23.3218	13.7876	345.439	89.1858	14.4766
6	24.652	13.6305	437.397	112.928	14.4766
7	25.9822	13.5296	526.057	135.818	14.4766
8	27.3123	13.4844	609.27	157.302	14.4766
9	28.6425	13.4947	685.145	176.891	14.4765
10	29.9727	13.5605	752.016	194.156	14.4765
11	31.3028	13.6822	809.093	208.893	14.4766
12	32.633	13.8604	855.789	220.948	14.4765
13	33.9632	14.0961	890.934	230.022	14.4765
14	35.2934	14.3906	913.513	235.852	14.4766
15	36.6235	14.7456	922.66	238.213	14.4765
16	37.9537	15.1632	917.649	236.92	14.4766
17	39.2839	15.646	897.9	231.821	14.4766
18	40.614	16.1974	862.979	222.805	14.4766
19	41.9442	16.8212	812.614	209.801	14.4765
20	43.2744	17.5224	746.712	192.787	14.4766
21	44.6046	18.3068	665.391	171.791	14.4765
22	45.9347	19.1821	569.028	146.912	14.4765
23	47.2649	20.1576	458.325	118.331	14.4766
24	48.5951	21.2456	335.375	86.5874	14.4765
25	49.9252	22.4619	211.06	54.4917	14.4766
26	51.2554	23.8282	92.2673	23.8217	14.4766
27	52.5856	25.3754	-13.7567	-3.55173	14.4766
28	53.9158	27.1493	-96.6994	-24.9659	14.4765
29	55.2459	29.2244	-140.69	-36.3235	14.4766
30	56.5761	31.7371	-119.43	-30.8346	14.4766
31	57.9063	35	0	0	0

Discharge Sections

Entity Information

Water Table

X	Y
0	15
18	15
30	23
48	29
66	32

External Boundary

X	Y
0	0
66	0
66	35
48	35
18	15
0	15